Medan EM 2.

No
Date

Bu Hepy

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Gradien potensial listrik digunakan u/ mengetahui potensial listrik keadaan potensial listrik minimum. maksimum.

digunakan untuk mengetahui.
potensial listrik di daerah tertentu itu konstan.

divergensi digunakan u/
u/ mengetahui di suatu tempat itu ada / tidak

Divergensi.
kerapatan Fluks listrik:

$$\nabla \cdot \hat{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$ (sistem koordinat kartesian)

$\hat{D}$ silinder $= D_\rho \hat{a_\rho} + D_\phi \hat{a_\phi} + D_z \hat{a_z}$ (sistem koordinat silinder).

$\hat{D}$ silinder : $D_\rho \hat{a_\rho}$

Contoh : $\hat{D} = 2 \sin\phi \; \hat{a_\phi}$ → $D_\rho = 0$, $D_\phi = 2\sin$, $D_z = 0$

$$\nabla \cdot \hat{D} = \frac{1}{\rho}\frac{\partial(\rho \cdot 0)}{\partial \rho} + \frac{1}{\rho}\frac{\partial(2\sin\phi)}{\partial \phi} + \frac{\partial(0)}{\partial z}$$

$$\nabla \cdot \hat{D} = 2\cos\phi.$$

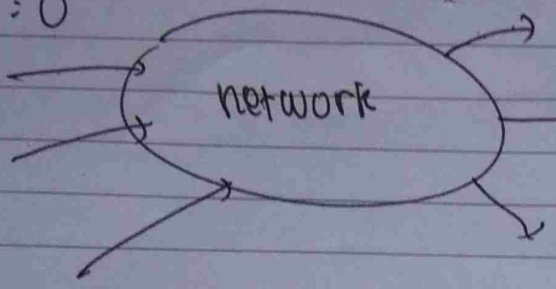
$$\nabla \cdot \hat{D} = \frac{1}{\rho}\frac{\partial(\rho D_\rho)}{\partial \rho} + \frac{1}{\rho}\frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

Materi

Tambahan

No
Date

$\nabla . \vec{D} = 0$



→ masuk = keluar
Tidak ada sumber apapun, tidak ada absorber, dan tidak ada / terjadi apa-apa.

Gradien

- diberlakukan saat beraran skalar hasil vektor.
  - memiliki arah + magnitudo
    - posisi max dari potensial listrik.

→ Jika gradien terhadap potensial listrik = 0 → maka potensial listrik konstan

→ $\vec{E}, \vec{H}, \vec{D}, \vec{B}$

Divergensi.

diberlakukan pada besaran vektor hasil skalar.
→ u/ mengetahui di suatu tempat terdapat sumber / tidak.

Sumber
- listrik
- magnetik → arus listrik → bergerak → menentukan arah.

Divergensi terhadap kerapatan fluks magnetik = 0.

$\nabla . \vec{D}$ → do dot del vektor.

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No
Date

Contoh:

$\vec{D} = 2y^3 z\, a\hat{x} + 3xy^3 z\, a_z$

$= 0 + 3xy^3$

$= 3xy^3$ //

$D_x = 2xy^2$

$D_y = 0$

$D_z = 0$

$\nabla \cdot \hat{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$

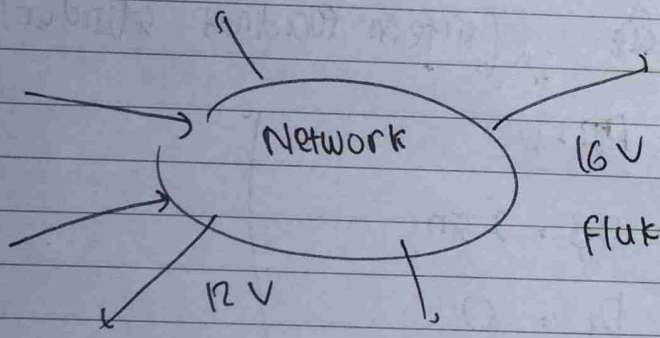
$\nabla \cdot \hat{D} = \frac{\partial(2xy^2)}{\partial x} + \frac{\partial(0)}{\partial y} + \frac{\partial(0)}{\partial z}$

$= 2y^3$ (skalar)

$\nabla \cdot \hat{D}$ misal di titik observator.
P (x = 2, y = 1, z = 3)

maka $\nabla \cdot \hat{D} = 2(1)^2 = 2$ //.

$\hat{D} = \rho^2 \sin \phi$



Fluks out.

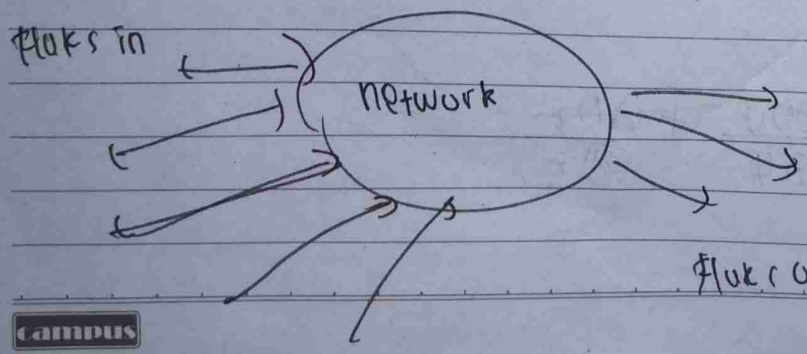
Fluks in.

Fluks keluar > Fluks masuk

Fluks out

$\nabla \cdot \hat{D} > 0$

Berarti ada sumber listrik di daerah tersebut.



Fluks out

$\nabla \cdot \hat{D} < 0$

ada absorber (penyerap gelombang) yang membuat fluks listrik menjadi lebih kecil dari fluks masuk.

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Contoh soal (Bola)

$\hat{D} = 2\sin\theta\sin\phi\,\hat{a}_\theta$

$\nabla \cdot \hat{D}$?

$D_\theta = 2\sin\theta\sin\phi$

$D_\phi = 0$

$D_r = 0$

$$\nabla \cdot \hat{D} = \frac{1}{r^2}\frac{\partial(r^2 D_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial(\sin\theta D_\theta)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial D_\phi}{\partial\phi}$$

$$= \frac{1}{r^2}\frac{\partial(0)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial(2\sin^2\theta\sin\phi)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial(0)}{\partial\phi}$$

$$= 0 + \frac{2\sin\phi}{r\sin\theta}\frac{\partial(\sin^2\theta)}{\partial\theta} + 0$$

$$= \frac{2\sin\phi}{r\sin\theta}(\sin 2\theta)$$

$$= \frac{2\sin 2\theta\sin\phi}{r\sin\theta} = \frac{2(2\sin\theta\cos\theta)\sin\phi}{r\sin\theta}$$

Absorber → banyak manfaatnya. salah satunya dipakai di badan pesawat tempur. agar tidak terdeteksi oleh musuh.



Divergensi fluks listrik = 0 adalah fluks in = fluks out. artinya tidak ada sel EM dan hanya lewat (tidak ada sumber dan tidak ada absorber).

1. Jika fluks in < fluks out, maka didalam black box terdapat sumber muatan listrik.
2. Jika fluks out < fluks in, maka didalam black box terdapat material penyerap (absorber).

No
Date

$\hat{D} = 2xy^2 a\hat{x} \Rightarrow$ $D_x = 2xy^2$

$D_z = 0$

$D_y = 0$

$\nabla . \hat{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$

$= \frac{\partial (2xy^2)}{\partial x} + \frac{\partial (0)}{\partial y} + \frac{\partial (0)}{\partial z}$

$= 2y^2$ (skalar).

~~K~~/. $\hat{D}$ = misal dititik observator
$P(x=2, y=1, z=3)$

maka $\nabla . \hat{D} = 2(1)^2 = 2$ //

$\hat{D} = 2y^2 x\, a\hat{x} + 3xy^2 z\, a\hat{z}$

Diketahui

$D_x = 2y^2 z$

$D_y = 0$

$D_z = 3xy^2 z$

$\nabla . \hat{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$

$= 0 + 0 + 3xy^3$

$= 3xy^3$ //

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$$\nabla \cdot \hat{D} = \frac{1}{r^2}\frac{\partial(r^2 D_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial(\sin\theta D_\theta)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial D_\phi}{\partial\phi}$$

(Sistem koordinat Bola)

$$\hat{D} = D_r \hat{a_r} + D_\theta \hat{a_\theta} + D_\phi \hat{a_\phi}$$ //

Lanjutan fluks

3. Jika fluks out = fluks in, maka di dalam block box tidak terdapat apa-apa.

Tugas

Latihan Soal

Tentukan Diverjens dari medan vektor berikut ini :

1. $\vec{D} = \vec{a_y}$ ⟶ $D_x = 0$ $D_y = 1$ $D_z = 0$

$$\nabla \cdot \vec{D} = \frac{\partial(0)}{\partial x} + \frac{\partial(1)}{\partial x} + \frac{\partial(0)}{\partial z}$$
$$= 0 + 0 + 0 = 0$$ //

2. $\vec{D} = x\vec{a_y}$

$D_x = 0$ $D_y = x$ $D_z = 0$

$$\nabla \cdot \vec{D} = \frac{\partial(0)}{\partial x} + \frac{\partial(x)}{\partial y} + \frac{\partial(0)}{\partial z}$$
$$= 0 + 0 + 0$$
$$= 0$$ //

3. $\vec{B} = x\vec{a}_x$

$\nabla \cdot \vec{B} = 0 \longrightarrow$ Diakarenakan divergensi terhadap kerapatan fluks listrik / magnetik = 0

4. $\vec{D}\ \frac{1}{\rho}\ \vec{a_\rho}$

$\nabla \cdot \vec{D} = \frac{1}{\rho}\frac{\partial(\rho \frac{1}{\rho})}{\partial \rho} + \frac{1}{\rho}\frac{\partial(0)}{\partial \phi} + \frac{\partial(0)}{\partial z}$

$D\rho = \frac{1}{\rho}$

$= \frac{1}{\rho} \cdot 0 + 0 + 0$

$D\phi = 0$

$= 0$ //

$Dz = 0$

5. $\vec{B} = \vec{a_r}$

$\nabla \cdot \vec{B} = 0 \longrightarrow$ Karena divergensi terhadap kerapatan fluks magnetik = 0

6. $\vec{D} = 3\rho\,\hat{a_\rho} + 6\hat{a_z}$

$D\rho = 3\rho$

$D\rho = 3\rho$

$D\phi = 0$

$Dz = 6$

$\nabla \cdot \vec{D} = \frac{1}{\rho}\frac{\partial(\rho \cdot 3\rho)}{\partial \rho} + \frac{1}{\rho}\frac{\partial(0)}{\partial \phi} + \frac{\partial(0)}{a_z}$

$= \frac{1}{\rho}\frac{\partial(3\rho^2)}{\partial \rho} + \frac{1}{\rho}\frac{\partial(0)}{\partial \phi} + \frac{\partial(0)}{a_z}$

$= \frac{1}{\rho}.(6\rho) + 0 + 0$

$= 6$ //